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Weather Pattern Visualization – Spatial-Temporal Analysis

Introduction :

Understanding meteorological variability across spatial and temporal dimensions is essential for climate monitoring, urban resource allocation, disaster risk mitigation, and agricultural planning. Atmospheric conditions fluctuate through localized geographical influences (such as latitude, elevation, and coastal proximity) as well as cyclical temporal factors (diurnal and seasonal patterns).

Spatial-temporal data science provides a rigorous framework to isolate macro-level climatic trends, map localized weather anomalies, and build predictive models for thermal shifts. By combining Python's data engineering and geospatial visualization ecosystem (Pandas, Folium, Plotly, Matplotlib, and Seaborn) with machine learning capabilities from Scikit-learn, this project models weather dynamics into interactive maps, correlation heatmaps, seasonal distributions, and time-series forecasting models.

Problem Statement :

The primary objective of this project is to perform a comprehensive spatial-temporal analysis on multi-city meteorological data to evaluate geographic climate variations and forecast temperature trends.

The dataset covers daily observations across eight major cities (New York, Los Angeles, Chicago, Miami, Seattle, Denver, Boston, and Phoenix) over a complete annual cycle (January 1, 2023, to December 31, 2023). Key objectives addressed include:

- Analyzing annual and seasonal temperature distributions across geographically distinct metropolitan areas.

- Mapping regional temperature tiers through interactive geospatial visualizations using Folium.
- Evaluating inter-feature dependencies among temperature, relative humidity, precipitation, wind speed, and atmospheric pressure using correlation matrices.
- Developing a supervised Linear Regression model to predict daily temperature patterns using temporal features, lagged variables, and atmospheric metrics.

Results and Discussion :

The spatial-temporal analysis and predictive modeling yielded the following core insights across the 2,920 meteorological records:

- **Geospatial & Seasonal Distribution:** Cities exhibited distinct thermal profiles based on geographical location. Southern and southwestern regions (Miami at 75.8°F, Phoenix at 74.4°F, and Los Angeles at 73.9°F mean temperature) maintained elevated temperatures throughout the year (>70°F tier), whereas northern and midwestern cities (Chicago at 45.0°F, Boston at 45.1°F, and Seattle at 45.2°F) experienced pronounced winter lows and high seasonal variance.
- **Precipitation & Moisture Trends:** Seasonal analysis highlighted strong localized precipitation dynamics, with coastal northwestern zones (Seattle) demonstrating elevated rainfall during late autumn and winter months, contrasting with summer-concentrated convective patterns in inland regions.
- **Correlation Dynamics:** Pearson correlation heatmaps revealed inverse relationships between temperature and barometric pressure during transition periods, alongside negative correlations between high heat and relative humidity in arid zones such as Phoenix.
- **Predictive Performance:** The Scikit-learn Linear Regression model successfully leveraged temporal indicators (day of the year, month, seasonal encoding) and lagged meteorological parameters to forecast daily temperatures with low Mean Squared Error (MSE) and a strong coefficient of determination (R^2).

(Insert your Folium interactive map screenshot, seasonal temperature trend line charts, monthly precipitation heatmaps, and regression evaluation metrics here)

Conclusion :

This project successfully developed an integrated spatial-temporal analysis and visualization pipeline for multi-city weather data. By combining data preprocessing,

feature engineering, interactive geospatial mapping with Folium/Plotly, and statistical evaluation, the study highlighted significant regional disparities in thermal stability and precipitation cycles. Furthermore, the implemented Linear Regression model proved effective in capturing baseline temperature trends from historical and atmospheric lag variables.

Future extensions of this work include incorporating real-time streaming data from global meteorological APIs, integrating deep learning architectures (such as LSTM networks or Spatial-Temporal Graph Neural Networks) for multi-step extreme event forecasting, and deploying the complete analysis as a cloud-hosted interactive dashboard using Streamlit.

References :

1. NOAA (National Oceanic and Atmospheric Administration) Climate and Weather Data Resources.
2. F. Pedregosa et al., *Scikit-learn: Machine Learning in Python*, Journal of Machine Learning Research (JMLR), 2011.
3. Folium & Leaflet.js Geospatial Mapping Documentation, Python Software Foundation, 2023.